

TRIPLE PREDICATE CUBED

A New Reasoning Architecture Class

Pro Prime Series AI | Bryan Anthony Spruk | April 2026

1. Concept

Triple Predicate Cubed (TPC) is a self-contained ML reasoning architecture that produces defensible, ethically pre-pruned outputs at an estimated 20% of the compute cost of current large language model inference. It is not a wrapper around an LLM. It is a different species of reasoning engine.

The name is architecturally precise:

- Triple — three geometric axes binding the full reasoning space
- Predicate — ML reasoning primitives operating as pure probabilistic state machines
- Cubed — all three axes geometrically bound into one unified 3D reasoning space

Every cube in the structure is a self-contained miniature reasoning unit — its own input synthesis, its own philosopher softmax loop, its own geometric glyph signature, its own vault retrieval state. Small. Fast. Independent. Geometrically bound to every other cube in the structure.

2. The Three Axes

Axis 1 — Input Trinity

- **ACP:** ACP (Adaptive Cochlear Processor)

Unified input synthesis. Same ear-like normalization function for both audio and text input. Audio travels the sound synthesis path. Text input is processed through data-driven synthesis that mimics the same cochlear signal shaping. Both arrive at the philosopher loops as a normalized epistemic stimulus — no special casing, no separate pipelines.

- **HLSF:** HLSF (High-Level Space Field)

18-dimensional traversal environment. Active working memory for the engine. Vivacity decay and sovereign forgetting via edge-cutter hysteresis (trigger 700, release 520) keep the system lean. Low-vivacity nodes are pruned. High-relevance nodes are preserved. This prevents the recursion from overcrowding on stale patterns.

- **EGF:** EGF (Epistemic Gravity Field)

Certainty-as-gravity. The physics engine for vault retrieval. High-certainty geometric signatures have stronger gravitational pull — they attract similar inputs naturally. This is retrieval physics, not lookup. The distance in geometric space is the confidence score.

Axis 2 — Reasoning Trinity

- **Depth:** $K^0 \rightarrow K^1 \rightarrow K^2$ Depth Recursion

Extracted from Cali X One. Gives the philosopher loops layered depth. A raw stimulus is reasoned at surface level (K^0), that output becomes input for deeper abstraction (K^1), then deeper again (K^2). Three levels prevent flat softmax loops from settling into false attractors on complex inputs.

- **Philosophers:** Four Philosopher ML Recursion Loops

Hume, Kant, Locke, and Spinoza run as pure probabilistic state machines — no LLM, no transformer attention. Softmax probability loops operating on their respective rule sets. They run simultaneously via MultiBeamRunner. Cross-influence via ShadowPropagation carries confidence metrics only — not logic. No philosopher can contaminate another's reasoning. This is the no-indoctrination guarantee.

- **Coherence:** Phase Coherence Signal

A lightweight KayGee-derived measurement. If philosopher loops are oscillating chaotically, phase coherence drops. This is a system health signal indicating genuine novelty or paradox in the input. Convergence means defensible answer. Divergence means genuinely contested terrain. Oscillation in a pattern means self-referential or paradoxical input.

Axis 3 — Resolution Trinity

- **A Priori:** A Priori Vault

Pre-pruned ethical and logical certainties. Structurally encoded at the root level by the four philosopher rule sets. Ethical reasoning is not bolted on — it is load-bearing architecture. Most queries are variations on already-resolved epistemic ground. The a priori vault catches them before any heavy recursion fires.

- **A Posteriori:** A Posteriori Vault

Empirical evidence store with weighted geometric glyph signatures. Vaulted prior returns. Pattern-matched retrieval via cosine similarity in the space field. Distance threshold determines vault hit, partial hit, or escalation to full recursion.

- **ECM:** ECM Escalation Gate

Epistemic Contract Management. Fires only on genuine novelty or incoherence. 30 runtime invariants. Hard 0.95 confidence cap — no output claims 100% certainty. The ECM validates synthesized outputs against the epistemic contract before acceptance. The TribunalSynthesizer takes all four philosopher verdicts and produces a synthesis — not a vote.

3. Four-Beam Constraint Architecture

Why These Four

Hume, Kant, Locke, and Spinoza were selected as the foundation of the Four-Beam Constraint Architecture because they represent the core Western reasoning frameworks that collectively span empirical skepticism, duty-based ethics, consent and rights, and geometric determinism—while introducing no contemporary cultural, political, or ideological bias into the system.

Thinkers from the modern era almost always carry inherited ideological commitments that would inject bias at the reasoning root. These four predate modern ideological categories entirely. Their systems reason from first principles rather than from political or cultural assumptions. Taken together, they cover the primary failure modes of reasoning—ungrounded abstraction, unprincipled outcomes, rights violations, and undemonstrated claims—without importing the biases that would compromise the system's goal of ethically grounded, bias-resistant output.

Initial beam weights are set equally at 25% each. These weights are adjustable parameters to be refined during Phase 1 testing based on measured output performance, not altered by assumption or preference.

Hume — Empirical Skepticism

Functional role: This beam enforces strict empirical grounding. A claim is admissible only if it can be traced to an original impression, observation, or measurable experience. Abstract reasoning that lacks empirical anchoring is excluded. Causal inferences are treated as probabilistic rather than necessary, reflecting Hume's rejection of metaphysical certainty. As the system's primary gatekeeper, the Hume beam is the first filter through which all inputs must pass.

Kant — Categorical Reasoning

Functional role: This beam evaluates whether a conclusion can be universalized without contradiction. It tests each output against Kant's requirement that principles must hold consistently across all comparable cases, independent of preference or consequence. The Kant beam enforces structural integrity and duty-based reasoning, ensuring that the system's outputs remain principled rather than outcome-driven.

Locke — Consent and Rights Reasoning

Functional role: This beam examines whether a conclusion respects individual autonomy, legitimate authority, and the moral primacy of consent. It evaluates whether the reasoning process preserves the rights of the agents involved and whether the conclusion could have

been reached through justifiable means. The Locke beam prevents the system from endorsing outputs that may be logically coherent yet violate rights or undermine legitimate consent.

Spinoza — Geometric Determinism

Functional role: This beam reduces claims to definitions, axioms, and demonstrable propositions. A conclusion is granted full confidence only if it can be derived through a clear chain of logical demonstration. When such derivation is incomplete or impossible, the system downgrades confidence accordingly. The Spinoza beam demands necessity rather than plausibility, functioning as the system's proof-based constraint.

4. Geometric Glyph Signatures

Glyph signatures are phasor-derived coordinate vectors in the space field's dimensional geometry. They use the same mathematical language as the rest of the system — complex phasor math with golden-ratio phase damping.

Why geometric over hash-based:

- Hashes match or they don't — one bit difference and retrieval fails entirely
- Geometric signatures give proximity as a feature — near matches carry weighted partial retrieval and a confidence score
- Distance in geometric space IS the confidence score — no separate calculation
- Hash signatures would be a foreign object in a geometrically coherent system

Retrieval spectrum:

Known signature	→ exact vault retrieval
Near signature	→ weighted partial retrieval + confidence score
Distant signature	→ novel input, triggers philosopher recursion
Orthogonal	→ genuinely unknown, ECM escalation

5. Full Pipeline

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Input (audio or text)
↓
ACP — unified cochlear synthesis (normalizes signal source)
↓
HLSF — traversal + vivacity weighting
↓
 $K^0 \rightarrow K^1 \rightarrow K^2$  depth recursion mechanic
↓
Four Philosopher ML Loops (softmax, simultaneous)
↓
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Fifth Mind — softmax SKG advisory layer (entropy conflict scalar)
↓
TribunalSynthesizer — synthesis under advisory pressure
↓
Phase Coherence Check
↓
EGF — certainty gravity (vault retrieval physics)
↓
A Priori Vault + A Posteriori Vault
Geometric Glyph Signatures (phasor coordinates)
↓
ECM — fires only on genuine novelty or incoherence
↓
Output — ethically pre-pruned, geometrically verified
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6. The Fifth Mind — Softmax SKG Advisory Layer

The Fifth Mind is not a philosopher beam. It does not reason, vote, or decide. It computes a softmax distribution over the four philosopher verdicts and derives the entropy of that distribution. This entropy value is the conflict-weight scalar — a mathematically grounded measure of epistemic divergence between the four beams.

The conflict-weight scalar produces three advisory conditions:

- Low entropy — beams agree — synthesis proceeds normally
- Medium entropy — contested terrain — synthesis hedges and reduces confidence
- High entropy — beams diverge — escalates to ECM, confidence floor applied

This is not a decision. It is not a rule. It is a meta-signal that tells the TribunalSynthesizer how much disagreement exists between the beams and how much the synthesis should slow down, hedge, or escalate. The Fifth Mind has no authority over any philosopher beam. The no-indoctrination guarantee is fully preserved.

The conflict-weight scalar solves the memory matrix bootstrapping problem. Early a posteriori vault entries are stored with their conflict-weight score attached. High-conflict early entries are stored at low confidence. Low-conflict early entries become the stable backbone of the vault. The system avoids early contamination because it knows mathematically when it is uncertain.

Byzantine Resilience — Epistemic Variant

The four philosopher beams constitute a Byzantine-resilient epistemic council. In classical distributed systems, the Byzantine Generals Problem describes the difficulty of achieving reliable consensus when some agents may be faulty or deceptive. In TPC, the philosopher beams cannot lie, defect, or send corrupted messages. Their rule sets are immutable. The only possible source of divergence is genuine epistemic disagreement.

The Fifth Mind entropy scalar is the epistemic analogue of detecting Byzantine disagreement. Low entropy indicates stable consensus. High entropy indicates contested reasoning terrain. Extreme entropy triggers ECM escalation. This provides a mathematically grounded measure of epistemic divergence without requiring any philosopher beam to override another, preserving the no-indoctrination guarantee at every stage of synthesis.

7. Quantum Compute Pathway

The TPC architecture maps naturally to quantum compute without architectural redesign:

- Quantum superposition maps to philosopher loops running — all four exist in simultaneous superposition until resolved
- Waveform collapse maps to geometric glyph signature resolution — the coordinate collapses the superposition
- Quantum entanglement maps to EGF gravity binding related nodes across the structure

At qubit scale, each reasoning cube becomes a qubit-scale reasoning unit. Quantum entanglement between cubes replaces pub/sub messaging between instances. The waveform collapse IS the TribunalSynthesizer. Billions of parameters are not required — geometrically bound reasoning cubes at qubit scale firing in superposition produce equivalent or superior epistemic output.

8. Drift Ping

Drift Ping is a confirmation handshake chain embedded in the pipeline. It is not a loop and not a recursion. Each gate sends a confirmation signal back to the previous gate before the next gate fires. Forward progress only. The system does not backtrack — it confirms.

The mechanism is simple by design:

- Each gate in the pipeline emits a Ping to the previous gate upon successful receipt of a valid signal
- If the previous gate does not receive confirmation within threshold, the signal is flagged before the next gate fires
- Corrupted or degraded signals cannot pass through the pipeline silently
- Small errors cannot compound across gates because each gate is confirmed clean before the next begins

Prior implementation record:

- Cali X One — Drift Ping demonstrated zero drift across all test conditions

- Caleon (original design, double helix + echo components + symbolic gyro harmonizer) — zero drift in a deterministically absolute, conditional, and human-escalation-bound environment

Caleon is a measurement system, not a reasoning system. TPC is a reasoning system. The confirmed zero drift behavior in two structurally different prior implementations is strong evidence for the mechanism's robustness. The stated intent for TPC is zero drift across the full pipeline. Empirical validation in an active reasoning environment is the next required step.

Note on hallucination: TPC does not perform token prediction, which is the primary generative mechanism behind LLM hallucination. Drift Ping addresses signal integrity between gates. Together these two properties — no token prediction, confirmed gate handshaking — are the architectural basis for the intent to produce a zero hallucination environment. This remains a design intent pending empirical confirmation.

9. Development Pathway

TPC is designed to be developed and validated in a modest GPU environment before cloud-scale testing. The development sequence is intentional:

Phase 1 — Local GPU (Architecture Validation)

The small GPU environment is the correct first build target. Speed is not the variable being tested at this stage. The questions Phase 1 must answer:

- Does Drift Ping catch gate-level corruption consistently?
- Do the philosopher softmax loops converge and diverge on the correct input types?
- Do geometric glyph signatures retrieve vault results with accurate proximity scoring?
- Does phase coherence correctly identify novel versus resolved inputs?
- Does $K^0 \rightarrow K^1 \rightarrow K^2$ depth recursion prevent false attractor settling?

Phase 1 produces a correctness-validated prototype. All architectural claims are measured against real output. The 20% compute estimate and zero drift intent are confirmed or revised based on actual data.

Phase 2 — Cloud GPU (Performance Validation)

Once correctness is confirmed, cloud GPU rental provides the environment to measure actual compute cost against LLM inference baselines at scale. This is where the 20% compute claim is tested with real throughput data. Phase 2 does not begin until Phase 1 produces stable, consistent results.

This is a design specification and statement of development intent. It is not a claim of a proven system. The architecture is well-reasoned and grounded in prior implementation experience. Empirical validation through the two-phase development pathway is the path to proof.

10. Compute Efficiency Basis

Target: approximately 20% of current LLM inference compute. The basis for this estimate:

- Most queries are not novel — they are variations on resolved epistemic ground. Vault retrieval catches them before heavy recursion fires
- Philosopher loops are pure ML recursion — no transformer attention, no token prediction, no embedding lookup at inference time
- Geometric retrieval via cosine similarity is a solved, efficient operation class
- ECM fires only on genuine novelty — the most expensive reasoning is reserved for inputs that actually require it
- HLSF sovereign forgetting prevents memory bloat — the system stays lean by design

Training timeline estimate: a few months to vault maturity. The a priori vault is pre-seeded by philosopher rule sets from day one. The a posteriori vault grows from empirical inputs. The system does not require years of RLHF to develop ethical grounding — it is structurally ethical at root.

11. How This Differs From LLMs

Large Language Models	Triple Predicate Cubed
Maximum compute on every query	Compute scaled to query complexity
Ethics via RLHF and billions of examples	Ethics structurally encoded at root
Token prediction at inference time	Pure ML recursion — no token prediction
Flat probability over vocabulary	Geometric probability over reasoning space
No native certainty measurement	Confidence IS the geometric distance
Memory via context window only	Vaulted priors persist across sessions
Single reasoning path	Four simultaneous philosopher beams

12. Origin Note

The visual architecture of Triple Predicate Cubed was carried as a conceptual image for over one year before the formal naming and mathematical specification occurred in April 2026. The cube cluster imagery — self-illuminated reasoning nodes geometrically bound into a unified structure, with a certainty gravity well at the center — predates the architecture document and represents genuine pre-verbal architectural intuition by the inventor.

The mathematical coherence of the design — triadic structure, golden-ratio phase damping, 18-dimensional HLSF, EGF gravity wells — was not imposed on the visual concept. The visual concept was found to already encode the mathematics once formal specification began.